

COMP8347

Internet Applications / Distributed Systems

Week 4 Class Activity (28.05)

Challenge: Fixing the Guestbook App

Individual Assignment — Report

1. Problem Overview

a) What was wrong in the provided Guestbook app

The starter project was a Django app (guestbook) that was intentionally broken in nine places. None of the four required features worked end to end. The defects were:

#	Location	Defect
1	forms.py	Meta.fields only had 'message', so display_name was never rendered or saved.
2	forms.py	clean_message() returned the raw value, allowing empty / whitespace-only messages.
3	views.py (add_entry)	Session counter was re-assigned to its current value instead of incremented.
4	views.py (add_entry)	Cookie pre-fill set initial for a display_name field the form did not have.
5	views.py (set_name_cookie)	Cookie written as 'display_name' but read as 'displayname' (key mismatch).
6	views.py (clear_name_cookie)	Deleted a non-existent key 'display_namex', leaving the real cookie.
7	entry_list.html	'Set my name cookie' link sent no name parameter, so the value was blank.
8	add_entry.html	POST form missing CSRF token, so submissions failed with HTTP 403.
9	add_entry.html	Template rendered form.display_name, a field the form did not define.

b) The required user features (in my own words)

- **Add an entry** — a visitor types a display name and a message and submits them.
- **List entries** — all messages are shown, each with its author and the time it was posted.
- **Remember the name** — a cookie stores the visitor's name so the form is pre-filled next time.
- **Count this session's entries** — the page shows how many entries the visitor added this session.

2. Implementation Summary (After Fix)

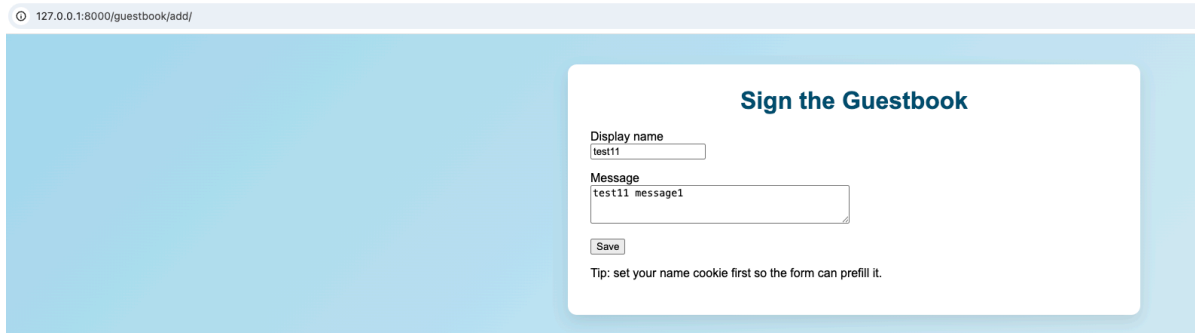
A. Add name + message

UI description. The 'Sign the Guestbook' page (/guestbook/add/) has a single-line Display name input and a multi-line Message textarea, plus a Save button. Each field shows a placeholder stating its maximum length (40 and 280 characters). The HTML maxlength attribute was deliberately removed so the user can type past the limit and the server-side validator can reject the input with a clear message.

What happens on submit. The form POSTs to add_entry. The view builds an EntryForm from request.POST; if it is valid it saves the Entry, increments the session counter, sets the name cookie, and redirects to the list (Post/Redirect/Get). If invalid, it re-renders the page with inline error messages.

Validation and error handling:

- **Empty name/message** — `clean_message()` strips whitespace and rejects empty input; the textarea is also required, so an empty message is never saved.
- **Overly long inputs** — limits are 40 characters for the name and 280 for the message (matching the model). Longer input is rejected server-side.



127.0.0.1:8000/guestbook/add/

Sign the Guestbook

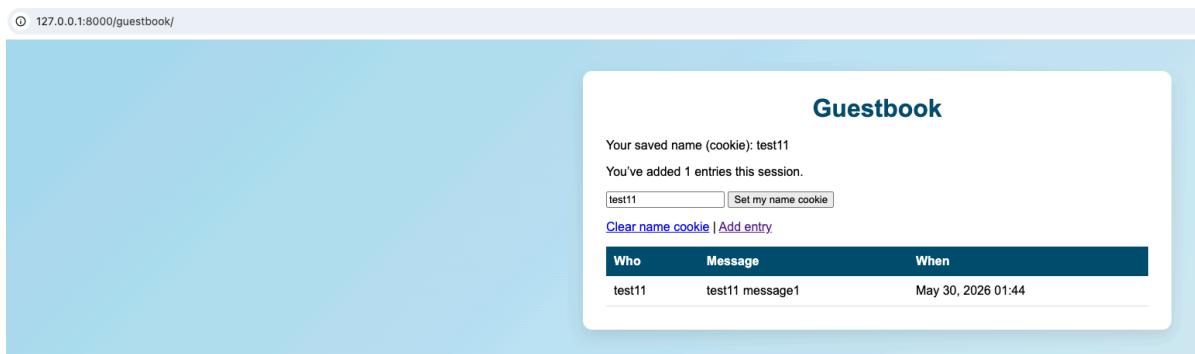
Display name
test11

Message
test11 message1

Save

Tip: set your name cookie first so the form can prefill it.

A1 — Add-entry form filled in (Display name + Message).



127.0.0.1:8000/guestbook/

Guestbook

Your saved name (cookie): test11

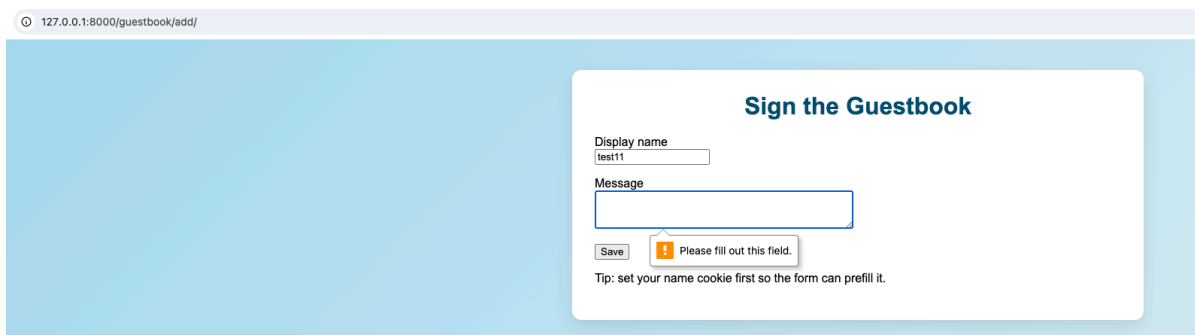
You've added 1 entries this session.

test11 Set my name cookie

[Clear name cookie](#) | [Add entry](#)

Who	Message	When
test11	test11 message1	May 30, 2026 01:44

A2 — Entry added successfully; session counter shows 1.



127.0.0.1:8000/guestbook/add/

Sign the Guestbook

Display name
test11

Message

Save

Please fill out this field.

Tip: set your name cookie first so the form can prefill it.

A3 — Empty message is blocked on submit.

127.0.0.1:8000/guestbook/add/

Sign the Guestbook

Display name
test11

Message
Advanced Database Topics, Advanced Systems Programming, and Networking & Data Security. The progra

Save

Tip: set your name cookie first so the form can prefill it.

A4 — Over-long message input.

B. List messages with time + author

Data model (guestbook/models.py, class Entry):

Field	Type	Notes
display_name	CharField(max_length=40, blank=True, default="")	author; 'Anonymous' if blank
message	TextField(max_length=280)	the message body
created_at	DateTimeField(auto_now_add=True)	timestamp set on creation

- **Storage** — entries are kept in the starter app's SQLite database (db.sqlite3) via the Django ORM.
- **Sorting order** — newest first (order_by('-created_at')); a guestbook should surface the latest activity at the top.
- **Time formatting** — Django filter date:"M d, Y H:i", e.g. May 30, 2026 01:46. TIME_ZONE='UTC' and USE_TZ=True, so times are in UTC.

127.0.0.1:8000/guestbook/

Guestbook

Your saved name (cookie): test11

You've added 3 entries this session.

test11 Set my name cookie

[Clear name cookie](#) | [Add entry](#)

Who	Message	When
test11	test11 message2	May 30, 2026 01:46
test11	A 16-month professional master's program focused on industry-ready software engineering. Coursework spans Advanced Software Engineering, Internet Applications & Distributed Systems, Advanced Database Topics, Advanced Systems Programming, and Networking & Data Security. The progra	May 30, 2026 01:45
test11	test11 message1	May 30, 2026 01:44

B1 — Multiple messages listed with author and time, newest first.

C. Cookie to remember the user's name

- **Cookie name / value** — 'displayname', storing the visitor's display name as a string.
- **When set/updated** — on every successful submission (using the posted name), and via the 'Set my name cookie' form on the list page.

- **Expiry policy** — persistent cookie, max_age = 7 days. Persistent (not session) because the goal is to remember the visitor across visits.
- **How the UI uses it** — on GET, the view passes the cookie as the form's initial value so the Display name field is auto-filled; the list page also shows the saved name.

The screenshot shows the Chrome DevTools Application tab. The left sidebar lists various storage areas, with 'Cookies' expanded for the URL 'http://127.0.0.1:8000'. The main pane displays a table of cookies.

Name	Value	D...	P...	E...	Si...	H...	S...	S...	P...	C
csrftoken	45oYW6w0ROjD...	1...	/	2...	41				Lax	
displayname	test11	1...	/	2...	17					
sessionid	vqnmia28188ng...	1...	/	2...	41	✓			Lax	

C1 — DevTools showing the 'displayname' cookie set after the first submit.

127.0.0.1:8000/guestbook/add/

Sign the Guestbook

Display name
test11

Message

Save

Tip: set your name cookie first so the form can prefill it.

C2 — Add form auto-filled with the remembered name.

D. Track how many entries added this session

- **Definition of 'session'** — a server-side session (Django session framework, keyed by the sessionid cookie).
- **Where the counter is stored** — the server session dictionary: `request.session['gb_added']`.
- **When it increments** — only on a successful POST, after `is_valid()` and `save()`: `session['gb_added'] = get('gb_added', 0) + 1`.
- **Refresh** — preserved (same session cookie).
- **New tab (same browser)** — continues (same session cookie sent).
- **New browser / incognito** — a fresh session, so the counter restarts at 0.

127.0.0.1:8000/guestbook/

Guestbook

Your saved name (cookie): test11

You've added 3 entries this session.

test11

[Clear name cookie](#) | [Add entry](#)

Who	Message	When
test11	test11 message2	May 30, 2026 01:46
test11	A 16-month professional master's program focused on industry-ready software engineering. Coursework spans Advanced Software Engineering, Internet Applications & Distributed Systems, Advanced Database Topics, Advanced Systems Programming, and Networking & Data Security. The progra	May 30, 2026 01:45
test11	test11 message1	May 30, 2026 01:44

D1 — Session entry counter.

127.0.0.1:8000/guestbook/

Guestbook

Your saved name (cookie): test11

You've added 4 entries this session.

[Clear name cookie](#) | [Add entry](#)

Who	Message	When
test11	test11 message3	May 30, 2026 01:50
test11	test11 message2	May 30, 2026 01:46
test11	A 16-month professional master's program focused on industry-ready software engineering. Coursework spans Advanced Software Engineering, Internet Applications & Distributed Systems, Advanced Database Topics, Advanced Systems Programming, and Networking & Data Security. The progra	May 30, 2026 01:45
test11	test11 message1	May 30, 2026 01:44

D2 — Counter increases as more entries are added.

3. Verification

An automated test suite (guestbook/tests.py, 5 tests) confirms the behaviour: adding an entry redirects, increments the counter and sets the name cookie; the list shows author + message, newest first; the add form pre-fills the name from the cookie; and empty or over-long messages are rejected and not saved. All 5 tests pass with: `python manage.py test guestbook`.